Library Book Management System

Problem Definition Create a Java program to manage a collection of books using an ArrayList. The system should allow adding new books, removing issued books, searching for a book by title, and displaying all available books. The project should highlight dynamic resizing and easy data handling of ArrayList.

1. Create a Book Class

```
class Book {
   private String title;
   private String author;
   public Book(String title, String author) {
      this.title = title;
      this.author = author;
   }
   public String getTitle() {
      return title;
   }
public String getAuthor() {
      return author;
   }
   public void displayBook() {
      System.out.println("Title: " + title + ", Author: " + author);
   }
}
```

2. Create Main Library Class

```
import java.util.ArrayList;
import java.util.Scanner;

public class LibraryManagement {
   public static void main(String[] args) {

      ArrayList<Book> library = new ArrayList<>();
```

```
Scanner sc = new Scanner(System.in);

int choice;

do {
    System.out.println("\n--- Library Menu ---");
    System.out.println("1. Add Book");
    System.out.println("2. Remove Book (Issue)");
    System.out.println("3. Search Book");
    System.out.println("4. Display All Books");
    System.out.println("5. Exit");
    System.out.print("Enter choice: ");
    choice = sc.nextInt();
    sc.nextLine(); // consume newline

    switch (choice) {

        case 1:
            System.out.print("Enter title: ");
            String title = sc.nextLine();

            System.out.print("Enter author: ");
            String author = sc.nextLine();

            library.add(new Book(title, author));
            System.out.println("Book added successfully!");
            break;

        case 2:
            System.out.print("Enter title to remove: ");
            String removeTitle = sc.nextLine();
```

```
    boolean removed = false;

    for (Book b : library) {
        if (b.getTitle().equalsIgnoreCase(removeTitle)) {
            library.remove(b);
            System.out.println("Book removed (issued).");
            removed = true;
            break;
        }
    }

    if (!removed) {
        System.out.println("Book not found.");
    }
    break;

case 3:
    System.out.print("Enter title to search: ");
    String searchTitle = sc.nextLine();

    boolean found = false;

    for (Book b : library) {
        if (b.getTitle().equalsIgnoreCase(searchTitle)) {
            b.displayBook();
            found = true;
        }
    }

    if (!found) {
```

```
                System.out.println("Book not found.");
            }
            break;

        case 4:
            if (library.isEmpty()) {
                System.out.println("No books available.");
            } else {
                for (Book b : library) {
                    b.displayBook();
                }
            }
            break;

        case 5:
            System.out.println("Exiting...");
            break;

        default:
            System.out.println("Invalid choice.");
        }

    } while (choice != 5);

    sc.close();
  }
}
```

Output:

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 1

Enter title: Java Basics

Enter author: James Gosling

Book added successfully!

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 1

Enter title: Data Structures

Enter author: Mark Allen Weiss

Book added successfully!

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 4

Title: Java Basics, Author: James Gosling

Title: Data Structures, Author: Mark Allen Weiss

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 3

Enter title to search: Java Basics

Title: Java Basics, Author: James Gosling

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 2

Enter title to remove: Java Basics

Book removed (issued).

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 4

Title: Data Structures, Author: Mark Allen Weiss

--- Library Menu ---

1. Add Book

2. Remove Book (Issue)

3. Search Book

4. Display All Books

5. Exit

Enter choice: 5

Exiting...